

1

Polytopes, Polyhedra, and Cones

In this lecture we prove some fundamental properties, in particular the equivalence of the two definitions of polytopes in Definition 0.1.

Of course, one could ask whether it is really necessary to go through these details, since the result is quite obvious anyway, and complete proofs are in the books [133] [252] [403] [484]. There are several good reasons. One is that we can give proofs that introduce important machinery (like Fourier-Motzkin elimination), which is useful for other purposes as well. It also yields a basic algorithmic tool to deal with polytopes. Additionally, these proofs provide geometric intuition, which we will need later. We will also see polarity appear in this context quite naturally, because we do two versions of Fourier-Motzkin, which are related by polarity. The “usual” approach is to do only one version, and prove the second half using polarity — this saves some work, but avoids the very interesting polar version. Finally, our proofs are (meant to be) easy and transparent, following simple geometric ideas through some elementary linear algebra, so they might even be fun. (There should be no crying in this lecture.)

1.1 The “Main Theorem”

However, to make sure that the pain level does not go below zero, we start with a few definitions. In the following, we work with two versions of polyhedra — in the course of this lecture we will see that they are mathematically (but not algorithmically!) equivalent. The two concepts

have also proved to be fundamental in a new field called “computational convexity”; see Gritzmann & Klee [242, 243].

The first concept, an $\mathcal{H}$ -polyhedron, denotes an intersection of closed halfspaces: a set $P \subseteq \mathbb{R}^d$ presented in the form

$$P = P(A, \mathbf{z}) = \{\mathbf{x} \in \mathbb{R}^d : A\mathbf{x} \leq \mathbf{z}\} \quad \text{for some } A \in \mathbb{R}^{m \times d}, \mathbf{z} \in \mathbb{R}^m.$$

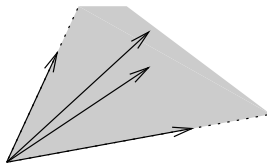
(Here “ $A\mathbf{x} \leq \mathbf{z}$ ” is the usual shorthand for a system of inequalities, namely $\mathbf{a}_1\mathbf{x} \leq z_1, \dots, \mathbf{a}_m\mathbf{x} \leq z_m$, where $\mathbf{a}_1, \dots, \mathbf{a}_m$ are the rows of A , and $z_1, \dots, z_m$ are the components of $\mathbf{z}$.)

For the second version we need the notion of a *cone*: a nonempty set of vectors $C \subseteq \mathbb{R}^d$ that with any finite set of vectors also contains all their linear combinations with nonnegative coefficients. In particular, every cone contains $\mathbf{0}$. For an arbitrary subset $Y \subseteq \mathbb{R}^d$, we define its *conical hull* (or *positive hull*) $\text{cone}(Y)$ as the intersection of all cones in $\mathbb{R}^d$ that contain Y . Clearly $C := \text{cone}(Y)$ is a cone for every Y . Similar to the situation for convex hulls (Lecture 0), one can easily see that

$$\text{cone}(Y) = \{\lambda_1 \mathbf{y}_1 + \dots + \lambda_k \mathbf{y}_k : \{\mathbf{y}_1, \dots, \mathbf{y}_k\} \subseteq Y, \lambda_i \geq 0\}.$$

In the case where $Y = \{\mathbf{y}_1, \dots, \mathbf{y}_n\} \subseteq \mathbb{R}^d$ is a finite set — this is the only case we will need here — this reduces to

$$\text{cone}(Y) := \{t_1 \mathbf{y}_1 + \dots + t_n \mathbf{y}_n : t_i \geq 0\} = \{Y\mathbf{t} : \mathbf{t} \geq \mathbf{0}\}.$$

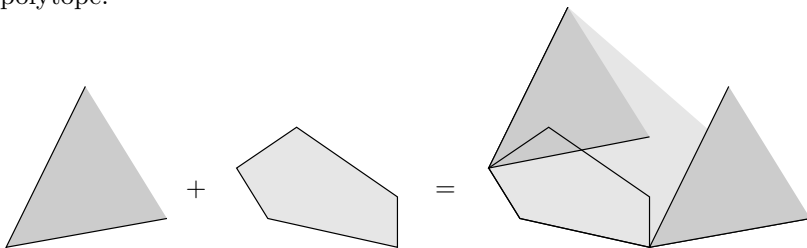


We define that $\text{cone}(Y) = \{\mathbf{0}\}$ if Y is the empty set, i.e., if $n = 0$.

The *vector sum* (or *Minkowski sum*) of two sets $P, Q \subseteq \mathbb{R}^d$ is defined to be

$$P + Q := \{\mathbf{x} + \mathbf{y} : \mathbf{x} \in P, \mathbf{y} \in Q\}.$$

The following sketch shows the 2-dimensional Minkowski sum of a cone and a polytope.



Now we define a $\mathcal{V}$ -polyhedron to denote any finitely generated convex-conical combination: a set $P \subseteq \mathbb{R}^d$ that is given in the form

$$P = \text{conv}(V) + \text{cone}(Y) \quad \text{for some } V \in \mathbb{R}^{d \times n}, Y \in \mathbb{R}^{d \times n'},$$

as the Minkowski sum of a convex hull of a finite point set and the cone generated by a finite set of vectors.

Thus, comparing this to Definition 0.1, we get that a $\mathcal{V}$ -polytope is a $\mathcal{V}$ -polyhedron that is *bounded*, that is, contains no ray $\{\mathbf{u} + t\mathbf{v} : t \geq 0\}$ with $\mathbf{v} \neq 0$. For this we only need to observe that $\text{conv}(V)$ is always bounded. This follows from a trivial computation: if $\mathbf{x} \in \text{conv}(V)$, then

$$\min\{v_{ki} : 1 \leq i \leq n\} \leq x_k \leq \max\{v_{ki} : 1 \leq i \leq n\},$$

which encloses $\text{conv}(V)$ in a bounded box. Similarly, an $\mathcal{H}$ -polytope is the same thing as a bounded $\mathcal{H}$ -polyhedron.

Now we start with a basic version of the “representation theorem for polytopes,” which will be considerably strengthened and generalized in the course of the proofs. See Section 2.4 for a definitive version.

Theorem 1.1 (Main theorem for polytopes).

A subset $P \subseteq \mathbb{R}^d$ is the convex hull of a finite point set (a $\mathcal{V}$ -polytope)

$$P = \text{conv}(V) \quad \text{for some } V \in \mathbb{R}^{d \times n}$$

if and only if it is a bounded intersection of halfspaces (an $\mathcal{H}$ -polytope)

$$P = P(A, \mathbf{z}) \quad \text{for some } A \in \mathbb{R}^{m \times d}, \mathbf{z} \in \mathbb{R}^m.$$

This result contains two implications, which are equally “geometrically clear” and nontrivial to prove, and which in a certain sense are equivalent.

Why is this theorem important? It provides two independent characterizations of polytopes that are of different power, depending on the problem we are studying. For example, consider the following four statements.

- Every intersection of a polytope with an affine subspace is a polytope.
- Every intersection of a polytope with a polyhedron is a polytope.
- The Minkowski sum of two polytopes is a polytope.
- Every projection of a polytope is a polytope.

The first two statements are trivial for a polytope presented in the form $P = P(A, \mathbf{z})$ (where the first is a special case of the second), but both are nontrivial for the convex hull of a finite set of points. Similarly the last two statements are easy to see for the convex hull of a finite point set, but are nontrivial for bounded intersections of halfspaces.

Theorem 1.1 is the version we really need, a very basic statement about polytopes; however, it is not the most straightforward version to prove. Therefore we generalize it to a theorem about polyhedra, due to Motzkin [414].

Theorem 1.2 (Main theorem for polyhedra).

A subset $P \subseteq \mathbb{R}^d$ is a sum of a convex hull of a finite set of points plus a conical combination of vectors (a $\mathcal{V}$ -polyhedron)

$$P = \text{conv}(V) + \text{cone}(Y) \quad \text{for some } V \in \mathbb{R}^{d \times n}, Y \in \mathbb{R}^{d \times n'}$$

if and only if is an intersection of closed halfspaces (an $\mathcal{H}$ -polyhedron)

$$P = P(A, \mathbf{z}) \quad \text{for some } A \in \mathbb{R}^{m \times d}, \mathbf{z} \in \mathbb{R}^m.$$

First note that Theorem 1.1 follows from Theorem 1.2 — we have already seen that polytopes are bounded polyhedra, in both the $\mathcal{V}$ - and the $\mathcal{H}$ -versions. $\square$

Theorem 1.2 can be proved directly, and the geometric idea for this is sketched in Section 1.2. However, fighting one's way through the formulas is quite strenuous, mainly because the points in $\text{conv}(V) + \text{cone}(Y)$ are hard to manipulate. It turns out that it is much easier to “homogenize”: we pass from affine d -space to linear $(d+1)$ -space; for this, we adjoin an extra coordinate (which we will take as the zeroeth coordinate in the following), mapping the point $\mathbf{x} \in \mathbb{R}^d$ to the vector $\begin{pmatrix} 1 \\ \mathbf{x} \end{pmatrix} \in \mathbb{R}^{d+1}$.

This reduces Theorem 1.2 to the special case where P is a cone, which can be proved more easily.

Theorem 1.3 (Main theorem for cones).

A cone $C \subseteq \mathbb{R}^d$ is a finitely generated combination of vectors

$$C = \text{cone}(Y) \quad \text{for some } Y \in \mathbb{R}^{d \times n}$$

if and only if it is a finite intersection of closed linear halfspaces

$$C = P(A, \mathbf{0}) \quad \text{for some } A \in \mathbb{R}^{m \times d}.$$

We will prove Theorem 1.3 in Section 1.3. In the following we will usually refer to the *polyhedral cones* characterized by Theorem 1.3 simply as “cones,” because the objects we consider are clearly polyhedra. Note that every cone C , by definition, contains the origin $\mathbf{0}$.

Let us see here why Theorem 1.2 follows from Theorem 1.3 by homogenization. For this, we associate with every polyhedron $P \subseteq \mathbb{R}^d$ a cone $C(P) \subseteq \mathbb{R}^{d+1}$, as follows.

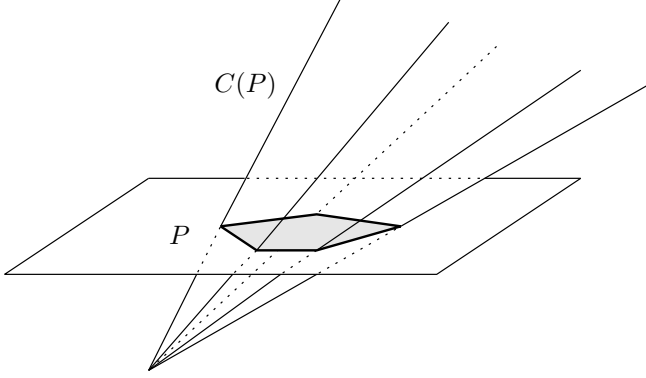
If $P = P(A, \mathbf{z})$ is an $\mathcal{H}$ -polyhedron, we define

$$C(P) := P\left(\begin{pmatrix} -1 & \mathbf{0} \\ -\mathbf{z} & A \end{pmatrix}, \begin{pmatrix} 0 \\ \mathbf{0} \end{pmatrix}\right).$$

That is, if P is defined by the inequalities $\mathbf{a}_i \mathbf{x} \leq z_i$, then $C(P)$ is defined by the inequalities $-z_i x_0 + \mathbf{a}_i \mathbf{x} \leq 0$, together with the inequality $x_0 \geq 0$. Clearly, $C(P)$ is again an $\mathcal{H}$ -polyhedron in $\mathbb{R}^{d+1}$, and

$$P = \{\mathbf{x} \in \mathbb{R}^d : \begin{pmatrix} 1 \\ \mathbf{x} \end{pmatrix} \in C(P)\}.$$

Also, we see that if $P = P(B, \mathbf{u})$ is an arbitrary $\mathcal{H}$ -polyhedron in $\mathbb{R}^{d+1}$, then $\{\mathbf{x} \in \mathbb{R}^d : \begin{pmatrix} 1 \\ \mathbf{x} \end{pmatrix} \in P\}$ is an $\mathcal{H}$ -polyhedron as well.



If $P = \text{conv}(V) + \text{cone}(Y)$ is a $\mathcal{V}$ -polyhedron, we define

$$C(P) := \text{cone} \begin{pmatrix} \mathbf{1} & \mathbf{0} \\ V & Y \end{pmatrix}.$$

Clearly, $C(P)$ is again a $\mathcal{V}$ -polyhedron in $\mathbb{R}^{d+1}$, and

$$P = \{\mathbf{x} \in \mathbb{R}^d : \begin{pmatrix} 1 \\ \mathbf{x} \end{pmatrix} \in C(P)\}.$$

Conversely, a simple computation shows that if $C = \text{cone}(W)$ is any cone in $\mathbb{R}^{d+1}$ generated by vectors $\mathbf{w}_i$ with $w_{i0} \geq 0$, then $\{\mathbf{x} \in \mathbb{R}^d : \begin{pmatrix} 1 \\ \mathbf{x} \end{pmatrix} \in C\}$ is a $\mathcal{V}$ -polyhedron.

Now, given any $\mathcal{H}$ -polyhedron P , we can apply Theorem 1.3 to $C(P)$, to conclude that $C(P)$ is a $\mathcal{V}$ -cone contained in $\{\mathbf{x} \in \mathbb{R}^{d+1} : x_0 \geq 0\}$, so P is a $\mathcal{V}$ -polyhedron as well. Conversely, if P is a $\mathcal{V}$ -polyhedron, then by Theorem 1.3 the associated cone $C(P)$ is an $\mathcal{H}$ -polyhedron, and hence so is P . $\square$

In both cases $C(P)$ realizes the *homogenization* of P , which we will discuss in Section 1.4, once we have established the Farkas lemma. The geometric idea is depicted in the sketch above, which shows the cone in $\mathbb{R}^3$ associated with an affine polytope in $\mathbb{R}^2$. If P is a polyhedron, then one has to add the necessary “points at infinity” to P , to make sure that $C(P)$ is a (closed) polyhedron.

1.2 Fourier-Motzkin Elimination: An Affine Sketch

For a direct proof of Theorem 1.2, the idea is the following. We have to see that every intersection of halfspaces ($\mathcal{H}$ -polyhedron) like $P(A, \mathbf{z})$ is a convex-conical combination ($\mathcal{V}$ -polyhedron) like $\text{conv}(V) + \text{cone}(Y)$, and conversely every convex-conical combination is an intersection of halfspaces.

For the forward direction (“every $\mathcal{V}$ -polyhedron is an $\mathcal{H}$ -polyhedron”) we note that every $\mathcal{V}$ -polyhedron

$$\begin{aligned} \text{conv}(V) + \text{cone}(Y) = \\ \{ \mathbf{x} \in \mathbb{R}^d : \exists \mathbf{t} \in \mathbb{R}^n, \mathbf{u} \in \mathbb{R}^{n'} : \mathbf{x} = V\mathbf{t} + Y\mathbf{u}, \mathbf{t} \geq 0, \mathbf{u} \geq 0, \mathbb{1}\mathbf{t} = 1 \} \end{aligned}$$

can be interpreted as the **projection** of a set

$$\{ (\mathbf{x}, \mathbf{t}, \mathbf{u}) \in \mathbb{R}^{d+n+n'} : \mathbf{x} = V\mathbf{t} + Y\mathbf{u}, \mathbf{t} \geq 0, \mathbf{u} \geq 0, \mathbb{1}\mathbf{t} = 1 \}$$

that quite clearly is an $\mathcal{H}$ -polyhedron. Thus it remains to show that

(I) any projection of an $\mathcal{H}$ -polyhedron is again an $\mathcal{H}$ -polyhedron.

This can be done by the *Fourier-Motzkin elimination* method: projecting down one dimension at a time. We will discuss only the case of projections along coordinate axes, which we need here; the general case can be reduced to this by an affine coordinate transformation.

In this section we give a geometric sketch for the case of affine polyhedra: the nice thing about it (as compared to the — more elegant — version for cones) is that its idea and most of its complications can already be illustrated in dimension 2, so we can provide pictures, and we will. However, instead of doing formulas for this version we switch to the version for cones — and do the proofs there.

We start with an $\mathcal{H}$ -polyhedron $P = P(A, \mathbf{z}) \subseteq \mathbb{R}^d$ and assume that we want to project to $\{ \mathbf{x} \in \mathbb{R}^d : x_k = 0 \} \equiv \mathbb{R}^{d-1}$ along the x_k -axis. The *projection* of $P \subseteq \mathbb{R}^d$ can be defined in great generality; we will only use the cases of coordinate directions, where we use the notation

$$\begin{aligned} \text{proj}_k(P) &:= \{ \mathbf{x} - x_k \mathbf{e}_k : \mathbf{x} \in P \} \\ &= \{ \mathbf{x} \in \mathbb{R}^d : x_k = 0, \exists y \in \mathbb{R} : \mathbf{x} + y \mathbf{e}_k \in P \}. \end{aligned}$$

for the *projection* of P in the direction of $\mathbf{e}_k$. The set $\text{proj}_k(P)$ is contained in the hyperplane $H_k = \{ \mathbf{x} \in \mathbb{R}^d : x_k = 0 \}$. A closely related set is the *elimination*

$$\begin{aligned} \text{elim}_k(P) &:= \{ \mathbf{x} - t \mathbf{e}_k : \mathbf{x} \in P, t \in \mathbb{R} \} \\ &= \{ \mathbf{x} \in \mathbb{R}^d : \exists y \in \mathbb{R} : \mathbf{x} + y \mathbf{e}_k \in P \}. \end{aligned}$$

Thus $\text{elim}_k(P)$ is the set of *all* points in $\mathbb{R}^d$ which project to $\text{proj}_k(P)$. In particular, we get an isomorphism $\text{elim}_k(P) \cong \text{proj}_k(P) \times \mathbb{R}$.